

COGS 566 - Probabilistic Models of Cognition

Final Report

Exploring Vagueness and Context-Sensitivity in Gradable Adjectives Using The Rational Speech Act (RSA) Framework

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Abstract

This project investigates four key aspects of gradable adjectives: (1) the nature of vagueness, (2) the occurrence of borderline cases, (3) the context-dependency of thresholds, and (4) the relationship between scale structure and the relative/absolute adjective distinction. The focus is on the positive form of gradable adjectives, analyzed through the lens of the Rational Speech Act (RSA) model. To explore these phenomena, synthetic datasets inspired by established experimental paradigms will be employed. The study aims to determine whether the RSA model can effectively approximate both to human agreement probabilities (synthetic data) and thresholds for gradable adjectives using synthetic linguistic datasets and also aims to determine whether it offers meaningful insights into the other identified phenomena.

Keywords: Gradable Adjectives, Relative Adjectives, Absolute Adjectives, Vagueness, Borderline Cases, Context-Dependency, Thresholds, Scale Structure, Rational Speech Act Model, Linguistic Semantics, Pragmatics.

1 Introduction

Adjectives are generally categorized into scalar (gradable) and non-scalar (non-gradable) types. Scalar adjectives are further divided into two subcategories: absolute and relative. Absolute adjectives can also be classified as total or partial (Burnett, 2014) [4]. Gradable adjectives can appear in comparative and superlative constructions and can be modified by intensifiers (Zhao, 2019) [6].

Gradable adjectives are most often studied in their positive form, which represents the morphologically simple and unmarked form. Examples include *tall* and *late*. In contrast, adjectives such as *atomic*, *pregnant*, and *geographical* are non-gradable (Burnett, 2014) [4], as they cannot be used in constructions like *This planet is more atomic than the other one*. The literature has extensively explored the distinction between absolute and relative adjectives in their positive form, examining how they can be differentiated based on gradable properties (degree scales) (Xiang et al., 2022 [7]; Qing Franke, 2014 [5]; Burnett, 2014 [4]). Additionally, literature also investigates how and why vagueness and context-sensitivity arise in these adjectives (Burnett, 2014 [4]; Kennedy, 2007 [3]; Lassiter & Goodman, 2013 [1]). These studies address

two primary questions: (1) What is the relationship between scale structure and the absolute/relative distinction? and (2) How does vagueness relate to contextual variability and borderline cases?

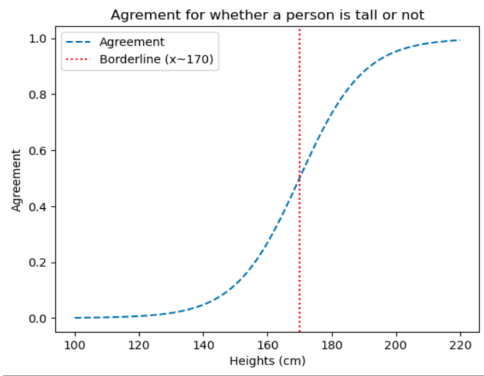
2 Dataset

The data for this project was generated using Python’s NumPy and SciPy libraries, drawing from distributions resembling the Sigmoid function. This approach ensures the data is both intuitively understandable and comparable to those found in the literature, such as in the study by Leffel et al. (2019) [2]. While Leffel et al. (2019) [2] collected data on the relative adjective *tall* from human participants—who rated their agreement with the adjective (and similar constructions) on a scale from 0 to 100 based on a person’s height—I synthesized data to approximate possible human agreement with the gradable adjective *empty*, besides the adjective *tall*. The synthetic data was designed to reflect agreement values drawn from distributions with meaningful threshold and steepness parameters, enabling an exploration of vagueness and context sensitivity in gradable adjectives in their positive form. To achieve this, distinct distributions were created for different adjectives, addressing the following research questions:

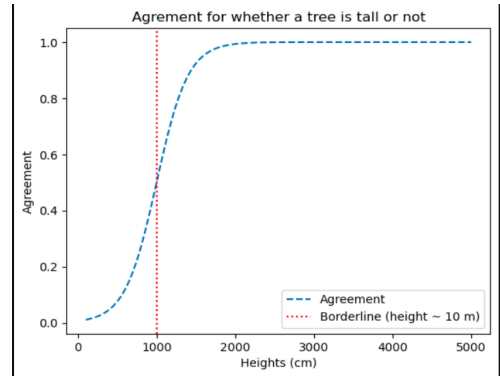
1. Is the gradable adjective vague?
2. Are there any borderline cases?
3. Are thresholds context-dependent?
4. Does the scale structure determine the relative/absolute distinction?

For conciseness, two examples are provided: the relative adjective *tall* and the absolute adjective *empty*. Addressing these questions requires at least two comparison classes for each adjective. For *tall*, **human** and **tree** serve as comparison classes, with heights measured in centimeters. For *empty*, **bucket** and **pool** are used as comparison classes, with the amount of liquid measured in liters.

The following plots illustrate the agreement distributions for the adjective *tall* based on the different values of human heights (Fig. 1a) and tree heights (Fig. 1b).

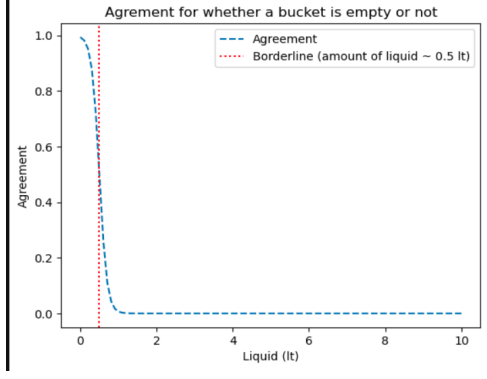


(a) First Comparison class (Human) for the relative adjective *tall*

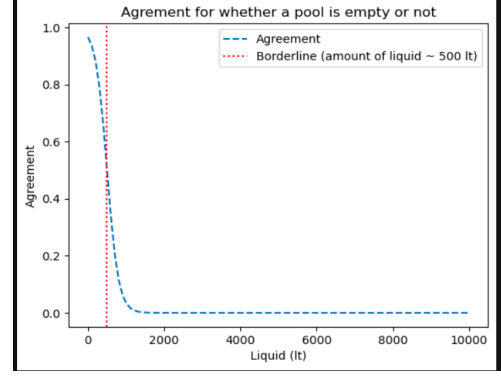


(b) Second Comparison class (Tree) for the relative adjective *tall*

The following plots illustrate the agreement distributions for the adjective *empty* based on the amount of liquid in a bucket (Fig. 2a) and the amount of liquid in a pool (Fig. 2b).



(a) First comparison class (Bucket) for the absolute adjective *empty*



(b) Second comparison class (Pool) for the absolute adjective *empty*

3 Model

Unlike the RSA models in the literature, this model can address questions about both agreements and thresholds. In order to study gradable adjectives, current RSA models typically focus on examining either agreements or thresholds separately. Moreover, this approach not only involves a novel (second) model but also requires new types of data. By leveraging the same datasets, this model can explore both agreements and thresholds through distinct meaning functions. In the original vanilla RSA, the meaning function assigns either 1 or 0 based on the respective probabilities. However, this model introduces a second meaning function, which also outputs 0 or 1 but employs a different calculation method while utilizing the same dataset.

The following plot (Fig. 3) illustrate the architecture of the model.

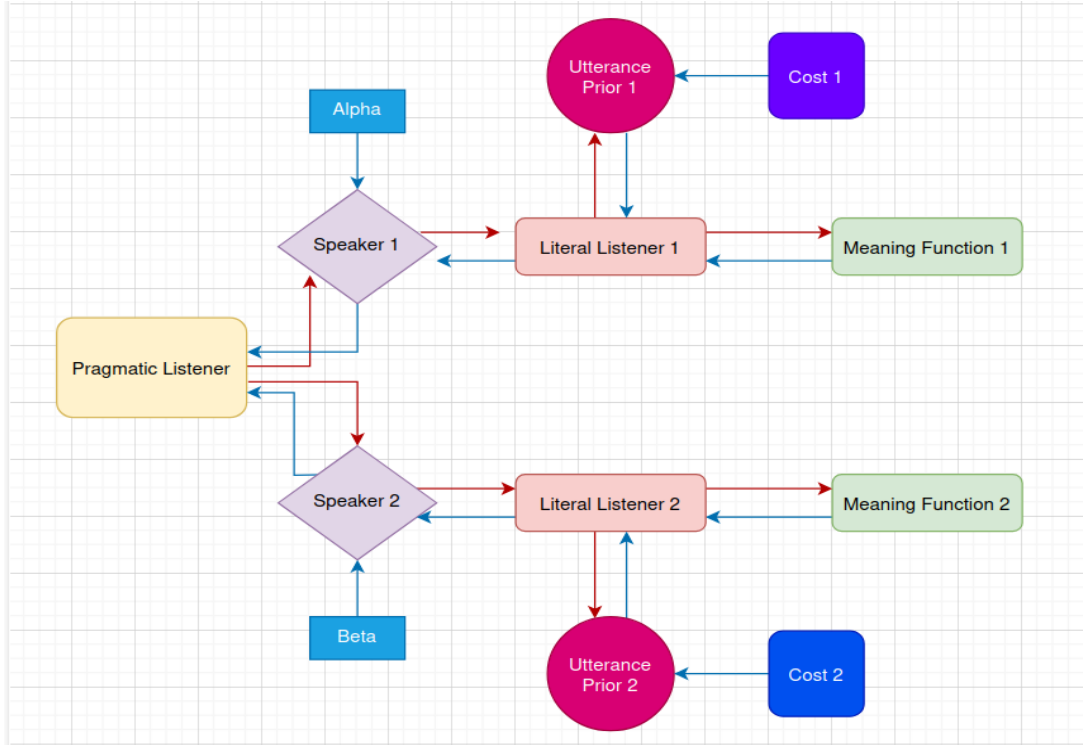


Figure 3: Model's Architecture

Meaning Function 1 : $[[u_1, agr]]^1 = flip(agr) \text{ if } u_1 = adj$

Utterances 1: $u_1 \rightarrow$ (Speaker 1 utters an adjective or stays silent.)

Literal Listener 1 : $L_{01} \rightarrow P_{L_{01}}(h | u_1, item) = P_{L_{01}}(h | [[u_1, agr]]^1 = 1)$

The input h is for measures like height for adjectives such as *tall*, it can also be amount of liquid for the adjective *empty*. The term *item* specifies whether the data belongs to first comparison class or second.

Meaning Function 2 : $[[u_2, agr]]^2 = flip(agr * (1 - agr)) \text{ if } u_2 = "yes"$

Utterances 2: $u_2 \rightarrow$ (Speaker 2 utters *yes* or stays silent.)

Literal Listener 2 : $L_{02} \rightarrow P_{L_{02}}(thr | u_2, item) = P_{L_{02}}(thr | [[u_2, agr]]^2 = 1)$
(The term *thr* is for threshold values and *agr* is the same data.)

Cost 1: $C_1 = adj : 1, "" : 0$

Speaker 1 : $P_{S_{01}} \rightarrow P_{S_{01}}(u_1 | h, item) \propto exp(\alpha \times [log(L_{01}(h | u_1, item)) - C_1(u_1)])$

Cost 2: $C_2 = "yes" : 1, "" : 0$

Speaker 2 : $P_{S_{02}} \rightarrow P_{S_{02}}(u_2 | thr, item) \propto exp(\beta \times [log(L_{02}(thr | u_2, item)) - C_2(u_2)])$

Pragmatic Listener: $P_{L_1} \rightarrow P_{L_1}(x, | q, u_x, item) \propto P_S(u_x | x, item) \times P_{L_1}(x)$

Here, q means query. It defines what pragmatic listener will do, therefore, what x will be. Then, it will call either speaker 1 or speaker 2. α and β are 2.5 except when comparison class is tree ($\alpha = 10$).

4 Model Outputs : Relative Adjective *tall*

The following plots (Fig. 4) illustrate the agreement of speaker 1 for utterancing the adjective *tall* based on the different heights of the five people.

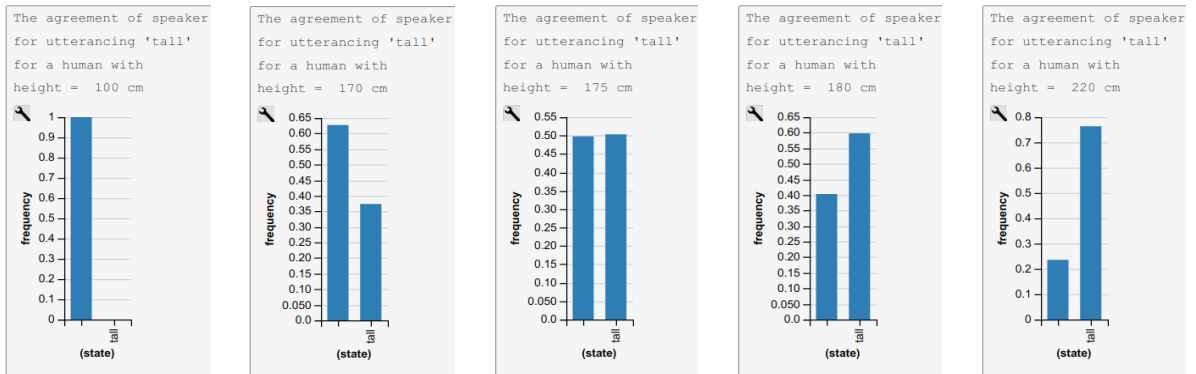


Figure 4: Agreement of speaker 1 for the utterance *tall* given different human heights.

The following plots (Fig. 5) illustrate the probability of speaker 2 for utterancing *yes* based on the different *threshold* values.

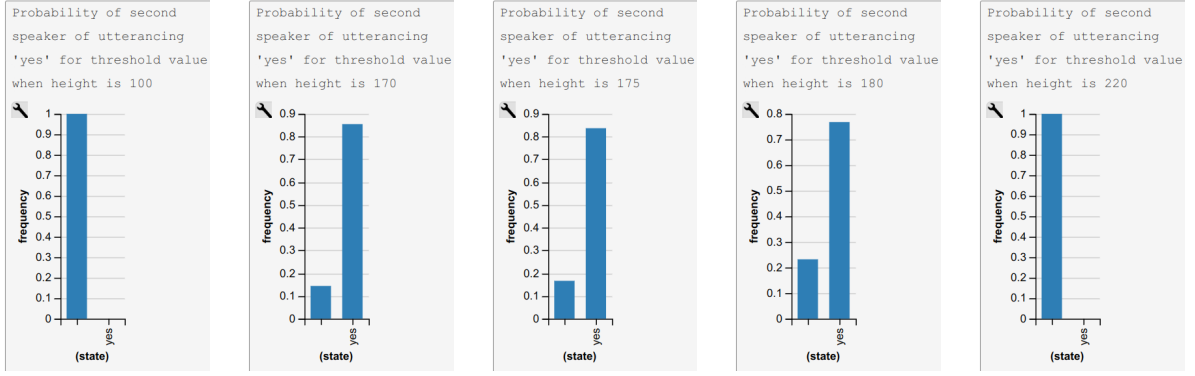


Figure 5: Probability of speaker 2 for the utterance *yes* given different human heights.

The following plots (Fig. 6) illustrate the agreement of speaker 1 for utterancing the adjective *tall* based on the different heights of the five trees.

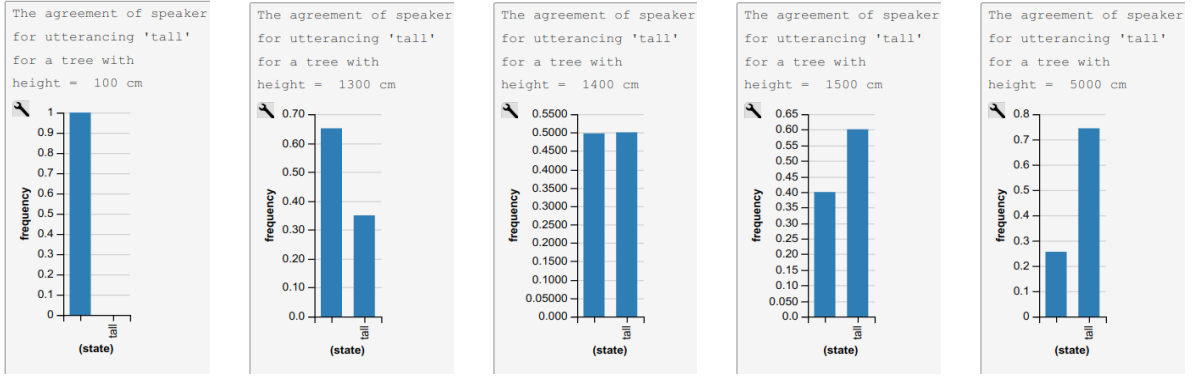


Figure 6: Agreement of speaker 1 for the utterance *tall* given different tree heights.

The following plots (Fig. 7) illustrate the probability of speaker 2 for utterancing *yes* based on the different *threshold* values.

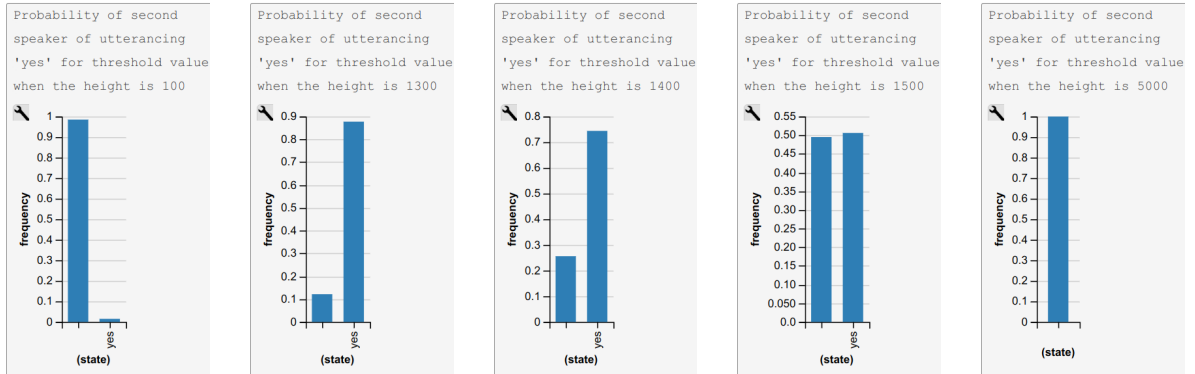


Figure 7: Probability of speaker 2 for the utterance *yes* given different tree heights.

5 Model Outputs : Absolute Adjective *empty*

The following plots illustrate the agreement of speaker 1 for utterancing the adjective *empty* based on the amount of liquid inside five buckets (Fig. 8).

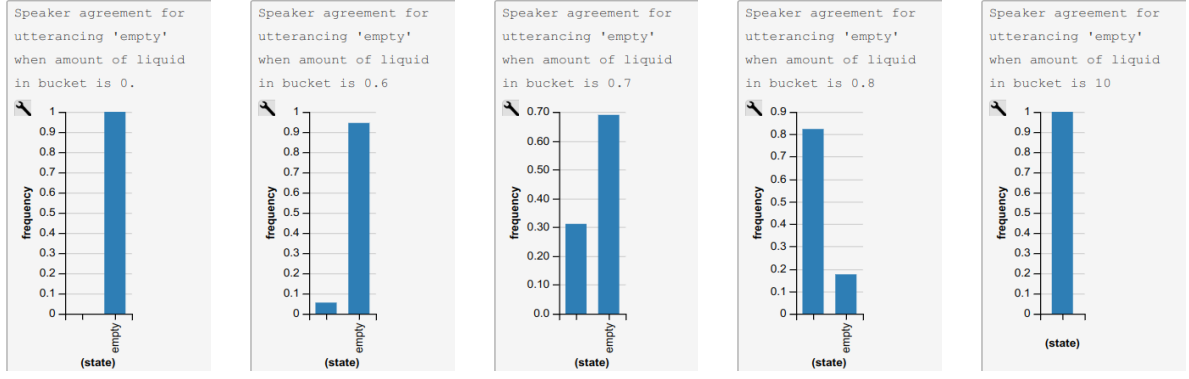


Figure 8: Speaker 1 agreements with the utterance *empty* for different amount of liquids in five buckets.

The following plots illustrate the probability of speaker 2 for utterancing *yes* for the thresholds based on the amount of liquids inside five buckets (Fig. 9).

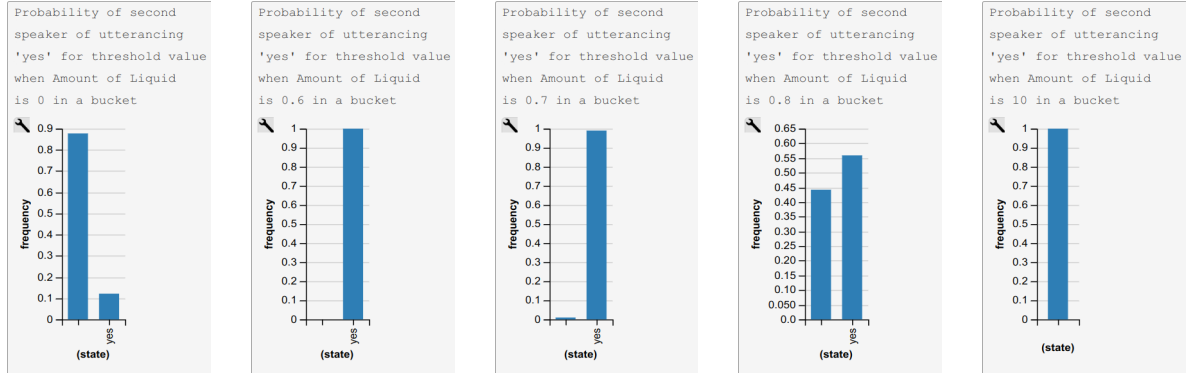


Figure 9: Speaker 2 agreements for the utterance *yes* given different amount of liquids in five buckets.

The following plots illustrate the agreement of speaker 1 for utterancing the adjective *empty* based on the amount of liquids inside five pools (Fig. 10).

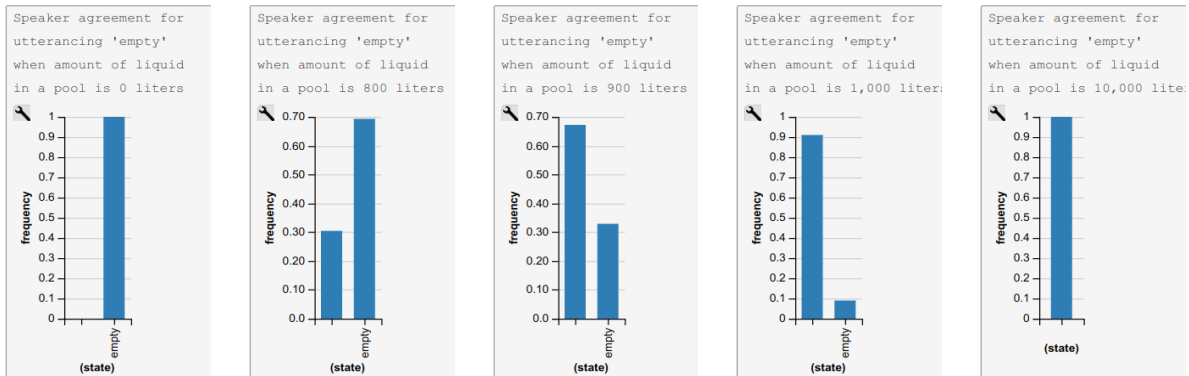


Figure 10: Agreement of speaker 1 with the utterance *empty* given different amount of liquids in five pools.

The following plots illustrate the probability of speaker 2 for utterancing *yes* for the thresholds based on the amount of liquids inside five pools (Fig. 11).

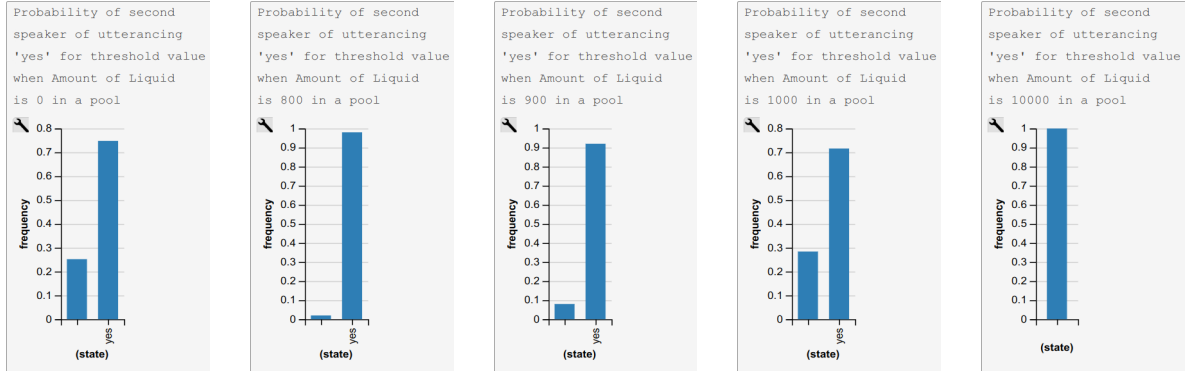


Figure 11: Speaker 2 agreements for the utterance *yes* given different amount of liquids in five pools.

The following plots illustrate the posterior probability distributions of pragmatic listener for the adjective *tall* and *empty* through the Speaker 1 given different comparison classes. (Fig. 12).

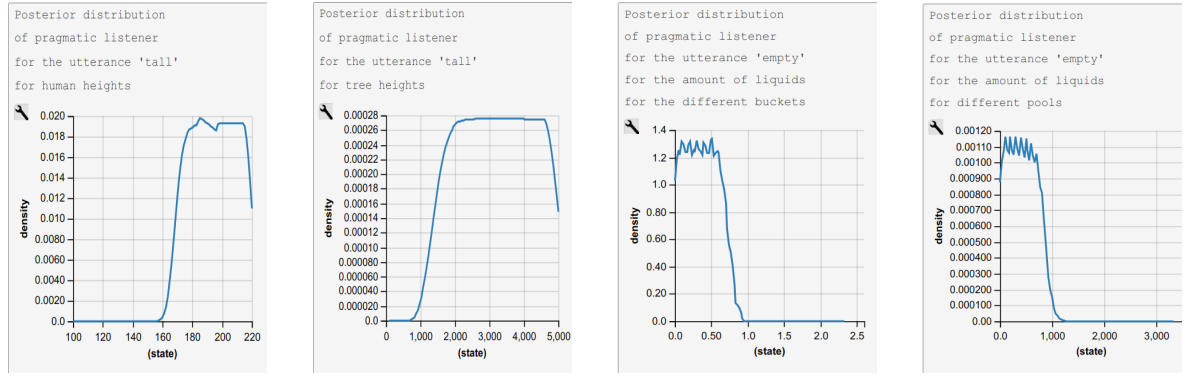


Figure 12: Posterior distributions of pragmatic listener for the adjective *tall* and *empty*

The following plots illustrate the posterior probability distributions of pragmatic listener for the thresholds for the adjective *tall* and *empty* via the Speaker 2 (Fig. 13).

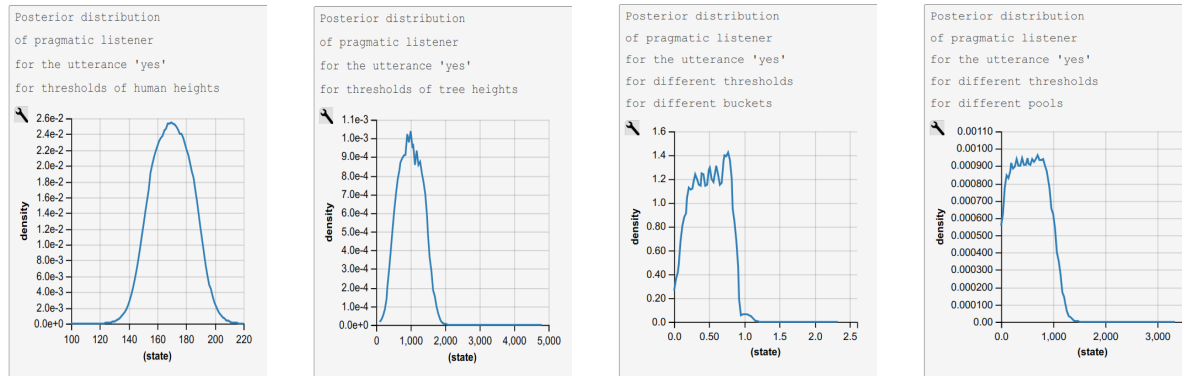


Figure 13: Posterior distributions of pragmatic listener for the thresholds for the adjective *tall* and *empty* through different comparison classes.

6 Conclusions :

6.1 Borderline Cases (regions) for the Relative Adjective *tall*

When we examine the plots above, it is evident that on the scale of tallness, there are height values where the probability (agreement) of the speaker uttering the adjective *tall* is close to 0.5. For humans, this value is approximately 175 cm, while for trees, it is around 1400 cm. These specific values can be considered **thresholds**. If there is a region surrounding the threshold with similar probabilities that lead to hesitancy in the speaker, these regions can be defined as **borderline regions**. This observation clearly demonstrates that the relative adjective *tall* has a borderline region, which is quite apparent from the data. As shown in the plot above, even if we did not choose the closest points to the threshold, the probabilities changed only slightly. The speaker can still hesitate about whether to utter *tall* or remain silent in those regions..

6.2 Borderline cases (regions) for the Absolute Adjective *empty*

First of all, there is no data point with a probability value close to 0.5, which could lead to hesitancy in the speaker, as we observed earlier for the relative adjective *tall*. In the **bucket** context, the threshold appears to be 0.7. However, even the threshold itself does not result in hesitancy.

The most important observation here is the abrupt changes in this region. When the observation shifts to the values closest to the threshold, 0.6 and 0.8, the hesitancy almost disappears. The speaker can confidently decide whether the bucket is *empty* or not and then produce the relevant utterance. This pattern is also observed in the other comparison class, **pool**.

As you may recall from the plots for the relative adjective *tall*, when the observation changed from the threshold to near values (they were not even the closest values), the change in agreement (probability) was very smooth, reflecting the existence of a clear borderline region. This is because relative adjectives like *tall* inherently have borderline regions. However, in this context, borderline regions are less apparent or may not exist at all for the absolute adjectives like *empty*.

6.3 Context-dependency of thresholds for the Relative Adjective *tall*

When analyzing the plots for speaker agreement for specific human and tree heights, it is clear that the threshold changes when the comparison class shifts from humans to trees and vice versa. For example, the threshold for the human comparison class is 175 cm, while for trees, it is 1400 cm. If we normalize these thresholds by dividing by the maximum values of their respective comparison classes, the threshold for humans becomes $175/220=0.80$, and for trees, it becomes $1400/5000=0.28$. This demonstrates that the threshold is context-dependent for the relative adjective *tall*. Similar examples can be found in the literature (e.g., Lassiter and Goodman, 2013) [1], confirming that thresholds for relative adjectives are context-dependent.

6.4 Context-dependency of thresholds for the Absolute Adjective *empty*

When analyzing the plots for speaker agreement regarding the adjective *empty* in the context of specific amounts of liquid in the **bucket** comparison class, the threshold lies near 0.7 liters. In contrast, in the **pool** context, the threshold is much higher, closer to 900 liters. If we normalize these data, for the **pool** context, $900 / 10000 = 0.09$, and for the **bucket** comparison class, $0.7 / 10 = 0.07$. Therefore, we can conclude that the threshold does not change significantly across comparison classes.

6.5 Vagueness for the Relative Adjective *tall*

As heights approach the threshold—within the borderline region—the speaker hesitates to decide whether to use the adjective *tall* or to remain silent. For instance, when a person’s height is 175 cm, the speaker’s agreement value for labeling this person as *tall* is approximately 0.5, indicating hesitation. Similarly, for tree heights, the same hesitation is evident; the speaker’s agreement value is close to 0.5 when observing a tree with a height of 1400 cm (14 m). This hesitation reflects the inherent vagueness of the relative adjective *tall*. Therefore, the relative adjective *tall* is vague due to the dependence of its meaning on context and the inherent ambiguity in borderline regions.

6.6 Vagueness for the Absolute Adjective *empty*

Unlike the relative adjective *tall*, the speaker does not hesitate when deciding whether to use the adjective *empty* or remain silent. For instance, even when the speaker observes the threshold value (0.7 liters for the bucket class), the probability of uttering *empty* is close to 0.7, indicating little or no hesitation, especially when compared to the relative adjective *tall*. Similarly, for the pool comparison class, the speaker can choose to stay silent with a probability close to 0.7 when observing a pool with 900 liters of liquid. Therefore, we can conclude that absolute adjective *empty* is less vague.

6.7 Scale Structure and Relative/Absolute Distinction

The scale structure plays a fundamental role in differentiating relative adjectives such as *tall* from absolute adjectives like *empty*, as evidenced by the provided plots. However, in the literature, these distinctions are mostly described in terms of threshold shifts and whether their scales are open or closed.

It is clear that relative and absolute adjectives exhibit fundamentally different behaviors for several reasons. For instance, relative adjectives show significant shifts in their thresholds depending on the comparison class, the emergence of clear borderline regions, and noticeable speaker hesitancy near those regions. Absolute adjectives, on the other hand, exhibit rapid transitions across thresholds and minimal speaker hesitancy, as seen in their plots. Therefore, we can draw the conclusion that scale structure determines relative and absolute distinction which means that we can classify them by examining their scales through different comparison classes.

References

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